

Electrostatic ballooning of spiders with multiple charged silk threads in near-surface turbulence: analytical limits and take-off probability

Alexandru Jicu,¹ Vlad Băutu,¹ and Octavian-Mihai Radu^{2,*}

¹*PoliPhysics Junior Lab, CAMPUS Research Institute,
National University of Science and Technology POLITEHNICA Bucharest, Bucharest, Romania*

²*AI Multimedia Lab, CAMPUS Research Institute,
National University of Science and Technology POLITEHNICA Bucharest, Bucharest, Romania*
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We start from the three-dimensional model of Habchi and Jawed [1], in which electrically charged silk threads are treated as discrete elastic rods subject to the atmospheric electric field, Coulomb repulsion, and air drag in the resistive force theory (RFT) approximation. We complete the model with the atmospheric field profile and experimentally measured charges [2, 3], and we place the bundle in synthetic near-surface turbulence generated by kinematic simulation with surface-layer statistics. Take-off becomes a stochastic event, characterized by a take-off probability; in the limit of eddies much larger and slower than the bundle, it reduces to a drift–diffusion result that smooths the deterministic threshold $\bar{F}_l = 1$ into a sigmoid controlled by a Péclet number. For tip-charged threads we obtain an exact first integral of the equilibrium equations, $T^\beta \sin \theta = \text{const.}$, and, in the small-angle limit, a closed-form expression for the opening of a bundle of N threads. The analytical results are compared with ensembles of numerical simulations for $N \leq 8$, which quantify the effect of finite bundle size, thread deformation, and Coulomb spreading on the take-off probability. Finally, we train a graph neural network surrogate with long-range message passing on the simulation data and show that it ? when tested on bundles with more threads than seen during training.

I. INTRODUCTION

Passive aerial dispersal of spiders (*ballooning*) is explained by two competing mechanisms: aerodynamic drag in rising air currents, and the electrostatic force exerted by the atmospheric potential gradient on negatively charged silk threads [2, 4]. From experiments in a chamber with a controlled electric field, Morley and Gorham estimated a total thread charge of about 1.15 nC for spiders of 0.9 mg [3]. Habchi and Jawed built the first three-dimensional model with multiple threads and showed that the Coulomb repulsion between threads produces a conical opening of the bundle [1]. Their model, however, assumes still air, although ballooning starts in the turbulent surface layer of the atmosphere.

The aerodynamic view has a long history. Humphrey [15] used a simple mechanical model of the spider–filament system to delimit, in terms of spider mass and silk length, the region in which ballooning is possible, and wind-tunnel experiments by Suter [16] showed that both the silk and the body contribute to the drag. Reynolds *et al.* [17] modelled the silk as a fully flexible, extensible line in turbulent flow and found that turbulence contorts the silk, so that ballooners have little control over their drag once airborne. More recently, Cho *et al.* observed large crab spiders (16–20 mg) taking off with 50–60 nanoscale fibres in weak updrafts of 0.1–0.5 m/s [7], and a bead–spring model with anisotropic drag was used to study the passive dynamics of a ballooning filament in shear, vortical, and turbulent flows [18]. None of these models includes electric charge. The electrostatic view received strong support when Morley and Robert showed that *Erigone* spiders detect vertical electric fields of atmospheric strength, that such fields alone elicit pre-ballooning behaviour and take-off in still air, and that the sensory hairs (trichobothria) respond mechanically to them [4]. Field-chamber experiments on three linyphiid species, however, found that wind had a much stronger effect on take-off than the electric field, the combination of the two giving the highest dispersal rates [9]; a recent review concludes that the relative importance of the two mechanisms is still open [10].

On the modelling side, thin elastic filaments are commonly simulated with the discrete elastic rods (DER) method [5], which represents a rod by its centreline nodes and edge twist angles and derives the forces from discrete stretching, bending, and twisting energies. Implicit time integration of DER makes very slender, nearly inextensible filaments tractable [1], and open-source implementations are available [6]. Habchi and Jawed coupled DER with resistive force theory and pairwise Coulomb repulsion, which is the starting point of the present work [1]. The cost

* ORCID: 0009-0002-1634-5065

of such simulations grows quickly with the number of threads, because the Coulomb interaction couples every pair of charged nodes and makes the Jacobian dense.

On the computational side, simulating many interacting charged filaments is expensive because the Coulomb interaction couples all nodes. Learned simulators based on graph neural networks reproduce the dynamics of particles at a fraction of the cost [14], and have been used to accelerate implicit solvers for rod dynamics [19], and long-range message-passing schemes such as Ewald message passing were designed precisely for $1/r$ interactions [13]. Whether such models generalize to filament bundles with more threads than seen during training has not been tested. **[TO DO: Octavian: add 2–3 references on GNN rod/cloth simulators and neural operators.]**

The contributions of this work are:

1. the multi-thread charged model of Habchi and Jawed placed in realistic synthetic near-surface turbulence, with take-off defined as a stochastic event;
2. an analytical drift–diffusion reference for the take-off probability and its comparison with ensemble simulations, which isolates the effect of the finite size and deformation of the bundle;
3. an exact first integral and a closed-form opening law for tip-charged thread bundles;
4. a graph neural network surrogate with long-range message passing, and a test of its generalization to unseen numbers of threads.

The paper is organized as follows. Section II presents the model. Section III describes the air flow and the take-off probability, and Sec. IV gives the analytical solution for tip-charged threads. Section V describes the numerical method and Sec. VI the learned surrogate. Results are presented in Sec. VII and discussed in Sec. VIII; Sec. IX concludes.

II. PHYSICAL MODEL

A. Geometry and parameters

The spider is a sphere of radius r_s , mass m , and charge Q_s , to which N identical threads of length L , radius r , and density ρ_t are attached. Each thread carries a charge q (in absolute value). As in [1], we consider two charge distributions: (a) charge concentrated at the thread tip, as suggested by [3]; (b) uniformly distributed charge, $\lambda = q/L$.

We use the following sign convention. The thread charge is negative and the fair-weather atmospheric field points downward, so the electrostatic force on the thread is *upward*. We work with the positive quantities q and $E(z)$; the force on the charge q is $+qE(z)\hat{z}$.

TABLE I. Reference parameters. Spider and thread values are those of [1].

Quantity	Symbol	Value	Source
Spider mass	m	1 mg (sweep 0.1–10 mg)	[1]
Spider radius	r_s	1 mm	[1]
Spider charge	Q_s	3 pC (negligible)	[1, 3]
Thread radius	r	300 nm	[1]
Thread length	L	0.5 m	[1, 3]
Silk density	ρ_t	1200 kg/m ³	[1]
Young’s modulus	Y	5–50 GPa (typical 25)	[1]
Air viscosity	μ	1.837×10^{-5} Pa s	
Field over flat ground	E_0	100–120 V/m	[2]
Field near a prominence	$E(z)$	Eq. (2), up to $\sim 10^5$ V/m	[1, 3]
Charge per thread	q	0.1–10 nC	[3]

B. Electric field

Over flat ground we use Gorham's profile [2]:

$$E(z) = E_0 e^{-\alpha z}, \quad E_0 = 120 \text{ V/m}, \quad \alpha = 3 \times 10^{-4} \text{ m}^{-1}. \quad (1)$$

On the scale of one thread ($L \sim 1 \text{ m}$), E is practically constant.

For the experiment of Morley and Gorham, with launch from an aluminum-foil-covered prominence, we use the fit given in [1]:

$$E(z) = E_1 e^{-z/z_1} + E_2 e^{-z/z_2} + E_\infty, \quad (2)$$

with $E_1 = 2.52 \times 10^5 \text{ V/m}$, $z_1 = 1.51 \text{ mm}$, $E_2 = 5.07 \times 10^4 \text{ V/m}$, $z_2 = 7.93 \text{ mm}$, and $E_\infty = 7.41 \times 10^3 \text{ V/m}$.

C. Forces

Let $s \in [0, L]$ be the arc length along a thread ($s = 0$ at the spider), $\mathbf{x}(s, t)$ the position, and $\mathbf{t} = \partial_s \mathbf{x}$ the unit tangent.

a. Air drag (RFT). For a very slender thread at low Reynolds number, the force per unit length is [1]

$$\mathbf{f}_v = -[\eta_{\parallel} \mathbf{t} \mathbf{t}^T + \eta_{\perp} (\mathbb{I} - \mathbf{t} \mathbf{t}^T)] (\mathbf{v} - \mathbf{u}), \quad (3)$$

$$\eta_{\parallel} = \frac{2\pi\mu}{\ln(L/r) - \frac{1}{2}}, \quad \eta_{\perp} = \frac{4\pi\mu}{\ln(L/r) + \frac{1}{2}}. \quad (4)$$

Here $\mathbf{v} = \partial_t \mathbf{x}$ is the thread velocity and $\mathbf{u}$ is the local air velocity. **The term $\mathbf{u}$ is our extension:** in [1], $\mathbf{u} = 0$. For the parameters of Table I, $\ln(L/r) \approx 14.3$, $\eta_{\parallel} \approx 8.35 \times 10^{-6} \text{ Pa s}$, and $\eta_{\perp} \approx 1.56 \times 10^{-5} \text{ Pa s}$. We define the ratio

$$\beta \equiv \frac{\eta_{\perp}}{\eta_{\parallel}} = \frac{2[\ln(L/r) - \frac{1}{2}]}{\ln(L/r) + \frac{1}{2}} \approx 1.87. \quad (5)$$

The spider body experiences Stokes drag, $\mathbf{F}_{v,s} = -6\pi\mu r_s (\mathbf{v}_s - \mathbf{u})$. We write $\zeta_s \equiv 6\pi\mu r_s \approx 3.5 \times 10^{-7} \text{ N s/m}$.

b. Electrostatic force. Per unit length, $\mathbf{f}_e = \lambda E(z) \hat{\mathbf{z}}$ for uniform charge; for tip charge, the force $qE(z) \hat{\mathbf{z}}$ acts at $s = L$.

c. Coulomb repulsion. Between any two charged thread elements $i \neq k$:

$$\mathbf{F}_{r,k} = k_e \sum_{i \neq k} \frac{q_i q_k}{\|\mathbf{x}_k - \mathbf{x}_i\|^3} (\mathbf{x}_k - \mathbf{x}_i), \quad k_e = \frac{1}{4\pi\epsilon_0}. \quad (6)$$

d. Weight. The thread weight is $N\rho_t \pi r^2 L g \approx 1.7 \times 10^{-9} \text{ N}$ for $N = 1$, about 10^{-4} of the spider weight. We neglect it in the analysis but keep it in the simulations.

D. Equilibrium equations for a flexible thread

Habchi and Jawed show that bending stiffness is negligible, with dimensionless ratio $\overline{YI} \sim 10^{-10}$ [1]. Each thread is a curved filament whose shape $\mathbf{x}(s)$ is part of the solution; straight segments appear only in the discretization of Sec. V. We therefore treat each thread as a perfectly flexible, inextensible string with tension $T(s)$, and neglect the (tiny) thread inertia. Local equilibrium reads

$$\partial_s (T \mathbf{t}) + \mathbf{f}_v + \mathbf{f}_e + \mathbf{f}_r = 0, \quad (7)$$

where $\mathbf{f}_r$ is the Coulomb force per unit length.

The boundary conditions are:

- at the spider ($s = 0$): $\sum_{j=1}^N T_j(0) \mathbf{t}_j(0) + (Q_s E - mg) \hat{\mathbf{z}} + \mathbf{F}_{v,s} = m \dot{\mathbf{v}}_s$;
- at the tip ($s = L$): for uniform charge, $T(L) = 0$; for tip charge, $T(L) \mathbf{t}(L) = qE \hat{\mathbf{z}} + \mathbf{F}_{r,\text{tip}}$.

For $N = 1$, $\lambda = 0$ and a horizontal wind, Eq. (7) reduces to problem A.1 of the physics competition, $\partial_s (T \mathbf{t}) = -\mathbf{f}_v$, with a drag force normal to the thread. Alex and Vlad must show this reduction explicitly; it is a good test of understanding.

E. Dimensionless groups

The global vertical balance for a nearly vertical bundle is

$$NqE - mg = (N\eta_{\parallel}L + \zeta_s)U, \quad (8)$$

where U is the vertical velocity of the spider *relative to the air*. This defines the characteristic velocity and the dimensionless groups:

$$\bar{F}_l = \frac{NqE}{mg} \quad \text{lift / weight (as in [1])}, \quad (9)$$

$$\Lambda = \frac{k_e q}{EL^2} = \frac{k_e q^2 / L^2}{qE} \quad \text{Coulomb repulsion / lift, per thread}, \quad (10)$$

$$\bar{w} = \frac{w(N\eta_{\parallel}L + \zeta_s)}{mg} \quad \text{dimensionless updraft}, \quad (11)$$

$$\beta = \eta_{\perp} / \eta_{\parallel}, \quad \bar{YI} = \frac{N^2 Y I}{k_e q^2} \quad (\text{negligible}). \quad (12)$$

[TO DO: Alex and Vlad verify with a dimensional analysis (Buckingham II theorem) that the set is complete and compute typical values for the parameters in Table I. For $q = 0.6$ nC, $E = 7.41 \times 10^3$ V/m, and $L = 0.5$ m they should obtain $\Lambda \approx 2.9 \times 10^{-3}$.]

III. AIR FLOW: MEAN-FLOW REFERENCE AND NEAR-SURFACE TURBULENCE

A. Uniform flow: a symmetry, not a new regime

In Eq. (3) the air velocity enters only through $\mathbf{v} - \mathbf{u}$. For a uniform and steady flow, $\mathbf{u} = \text{const.}$, the change of frame $\mathbf{x}' = \mathbf{x} - \mathbf{u}t$ maps the problem onto still air: accelerations are unchanged, and E varies negligibly over the distance travelled. The mapping is exact at all times if the initial state is the same relative to the air ($\mathbf{v}'(0) = \mathbf{v}(0) - \mathbf{u}$); otherwise the two motions differ during a transient and converge to the same steady state. The shape of the bundle and the velocity U relative to the air are therefore exactly those in still air, and the ground-frame velocity is $V = U + w$ for $\mathbf{u} = w\hat{z}$. This symmetry is used only as a test of the code (Fig. 8). A flow can change the shape of the bundle only if it varies in space or in time on scales comparable to L or to the response time of the spider.

B. Mean-flow reference: deterministic threshold

For a steady updraft w , the spider rises if $V > 0$. From Eq. (8),

$$V = w_s(\bar{F}_l - 1) + w > 0 \quad \Longleftrightarrow \quad w > w_c \equiv w_s(1 - \bar{F}_l), \quad w_s \equiv \frac{mg}{N\eta_{\parallel}L + \zeta_s}, \quad (13)$$

where w_s is the settling speed of the uncharged spider in calm air. In dimensionless form, $\bar{w}_c = 1 - \bar{F}_l$. This straight line is the reference against which the turbulent results of Sec. III C are compared. It assumes nearly vertical threads; Coulomb tilting of the threads (large Λ , large N) increases the drag and shifts the true boundary.

C. Near-surface turbulence

a. Why turbulence, and at which scale. Ballooning starts within a few metres of the ground, in the atmospheric surface layer. There, the standard deviation of the vertical velocity scales with the friction velocity, $\sigma_w \approx 1.25 u_*$ in neutral conditions [20, 21], and the dominant eddies have a size of the order of the height above ground, $\ell_w \sim z$ [20]. At take-off heights of 0.5–2 m this is comparable to the thread length $L = 0.5$ m. A realistic flow must therefore vary in space across the bundle, not only in time; a spatially uniform gust $w(t)$ would be artificial.

TABLE II. Mean-flow threshold w_c (m/s) from Eq. (13). $L = 0.5$ m, $r = 300$ nm, $E = 7.41 \times 10^3$ V/m, tip charge. Negative values: the spider rises even in a downdraft. Values above ~ 1 m/s exceed the range of validity of the low-Reynolds-number drag laws and should be flagged. The column $q = 0$ equals the settling speed w_s . Numerical values: $-V$ at steady state in still air ($Q_s = 0$, $N_t = 100$). The charged multi-thread cases lie up to 12% below the analytical value (in absolute value) because the Coulomb splay of the threads increases the drag. **[TO DO: Rows $m = 10$ mg: rerun with $t_{\text{end}} \geq 40$ s (current runs are not fully relaxed, $t_s \approx 2$ s).]**

N	m (mg)	$q = 0$		$q = 0.6$ nC per thread	
		analytical	numerical	analytical	numerical
1	0.1	0.217	0.218	-0.767	-0.770
1	1	2.17	2.18	1.19	1.19
1	10	21.7	?	20.7	?
2	0.1	0.113	0.114	-0.910	-0.882
2	1	1.13	1.13	0.106	0.106
2	10	11.3	?	10.3	?
4	0.1	0.058	0.058	-0.986	-0.909
4	1	0.576	0.579	-0.468	-0.458
4	10	5.76	?	4.71	?
8	0.1	0.029	0.030	-1.03	-0.898
8	1	0.291	0.293	-0.763	-0.708
8	10	2.91	?	1.85	?

b. Synthetic turbulence. We generate the air velocity by kinematic simulation [22, 23], a sum of N_k random Fourier modes,

$$\mathbf{u}(\mathbf{x}, t) = U_h \hat{\mathbf{x}} + \sum_{n=1}^{N_k} [\mathbf{a}_n \cos(\mathbf{k}_n \cdot \mathbf{x}' + \omega_n t) + \mathbf{b}_n \sin(\mathbf{k}_n \cdot \mathbf{x}' + \omega_n t)], \quad \mathbf{x}' = \mathbf{x} - U_h t \hat{\mathbf{x}}, \quad (14)$$

with random directions $\hat{\mathbf{k}}_n$, amplitudes $\mathbf{a}_n, \mathbf{b}_n \perp \mathbf{k}_n$ (so that $\nabla \cdot \mathbf{u} = 0$), and magnitudes set by the energy spectrum in the shell Δk_n around k_n , normalized so that $\frac{1}{2} \langle |\mathbf{u} - U_h \hat{\mathbf{x}}|^2 \rangle = \int_0^\infty E(k) dk$ [23]. The energy spectrum is of von Kármán form,

$$E(k) \propto \sigma^2 \ell \frac{(k\ell)^4}{[1 + (k\ell)^2]^{17/6}}, \quad \int_0^\infty E(k) dk = \frac{3}{2} \sigma^2, \quad (15)$$

which has the $k^{-5/3}$ inertial range. The mean horizontal wind U_h advects the eddies past the spider (Taylor's hypothesis); by the symmetry above it has no other effect. The unsteadiness frequencies are $\omega_n = \lambda \sqrt{k_n^3 E(k_n)}$ with $\lambda \approx 0.5$ [23]. Each random seed defines one realization; the modes are stored with the trajectory, so that each run can be reproduced exactly.

TABLE III. Turbulence parameters. Surface-layer scalings from [20, 21].

Quantity	Symbol	Range	Note
Friction velocity	u_*	0.1–0.3 m/s	light to moderate wind
Vertical velocity std. dev.	σ_w	$1.25 u_* \approx 0.12$ –0.38 m/s	neutral surface layer
Integral length scale	ℓ	0.5–2 m	$\sim$ height above ground
Mean horizontal wind	U_h	0.5–3 m/s	1 m/s in the simulations
Number of modes	N_k	200	$k_{\min} = 0.1/\ell$, $k_{\max} = 2\pi/(5\Delta\ell)$
Unsteadiness parameter	λ	0.5	[23]

c. Validity. Relative velocities are of order $\sigma_w \lesssim 0.4$ m/s, so the thread Reynolds number is $\text{Re} = \rho_{\text{air}} \sigma_w r / \mu \sim 10^{-2}$ and RFT remains valid. The model (14) is homogeneous and isotropic, whereas surface-layer turbulence is anisotropic ($\sigma_u > \sigma_w$) and its scale grows with height. We match the vertical statistics, $\sigma = \sigma_w$ and $\ell = \ell_w$, which control take-off. **[TO DO: Alex and Vlad: check numerically that the generated field has the prescribed σ_w and spectrum (Fig. 5).]**

D. Take-off probability

a. Definition. The spider is released at height z_0 above the ground (tip of a grass blade or twig) and followed until it either reaches a height h (*take-off*) or returns to the ground, $z = 0$ (*failure*). The take-off probability P is the fraction of realizations that reach h . It depends on $\bar{F}_l$, N , Λ , and the turbulence through σ_w/w_s and ℓ/L .

b. Analytical reference: drift-diffusion limit. If the eddies are much larger than the bundle ($\ell \gg L$) and their turnover time ℓ/σ_w is much longer than the response time of the spider,

$$t_s = \frac{m}{N\eta_{\parallel}L + \zeta_s} = \frac{w_s}{g} \approx 0.2 \text{ s} \quad (m = 1 \text{ mg}, N = 1), \quad (16)$$

the spider simply follows the local air with an added drift $U_0 = w_s(\bar{F}_l - 1)$. On times longer than the Lagrangian time $T_L \sim \ell/\sigma_w$, its height then performs a drift-diffusion with diffusivity $K = \sigma_w^2 T_L \sim \sigma_w \ell$ [24]. The probability to reach h before 0, starting from z_0 , solves $U_0 P' + K P'' = 0$ with $P(0) = 0$, $P(h) = 1$:

$$P = \frac{1 - e^{-U_0 z_0/K}}{1 - e^{-U_0 h/K}}, \quad U_0 = w_s(\bar{F}_l - 1), \quad K \simeq \sigma_w \ell \quad (17)$$

For $U_0 \rightarrow 0$, $P \rightarrow z_0/h$; for $|U_0| h/K \gg 1$, P switches sharply from 0 to 1 at $\bar{F}_l = 1$, recovering the deterministic threshold. Turbulence thus smooths the step $\bar{w}_c = 1 - \bar{F}_l$ into a sigmoid of width set by the Péclet number $\text{Pe} = w_s h/K$.

Equation (17) is a reference limit, not the answer. At take-off $\ell \sim L$ and eddy times $\sim \ell/\sigma_w \sim 1\text{--}5$ s are only one order of magnitude above t_s , so the assumptions behind it are only marginally satisfied. The deviation of the simulated P from Eq. (17), due to finite bundle size, thread deformation, and Coulomb spreading, is the main numerical result of the paper. Alex and Vlad must derive Eq. (17) independently.

[TO DO: Alex and Vlad: compute t_s , w_s , K , and Pe for the parameter ranges of Tables I and III.]

IV. ANALYTICAL SOLUTION FOR TIP-CHARGED THREADS

A. Equations along one thread

Consider the steady state with charge concentrated at the tip, so $\lambda = 0$ and $\mathbf{f}_r = 0$ along the thread. One thread lies in a vertical plane, at angle $\theta(s)$ from the vertical, with $\mathbf{t} = (\sin \theta, \cos \theta)$. In the spider frame the thread moves relative to the air with velocity $U\hat{\mathbf{z}}$, so $\mathbf{v} - \mathbf{u} = U\hat{\mathbf{z}}$.

We project on the tangent $\mathbf{t}$ and on the normal $\mathbf{n} = (\cos \theta, -\sin \theta)$, for which $\partial_s \mathbf{t} = \theta' \mathbf{n}$. Since $\hat{\mathbf{z}} \cdot \mathbf{t} = \cos \theta$ and $\hat{\mathbf{z}} \cdot \mathbf{n} = -\sin \theta$, Eq. (7) becomes

$$\frac{dT}{ds} = \eta_{\parallel} U \cos \theta, \quad (18)$$

$$T \frac{d\theta}{ds} = -\eta_{\perp} U \sin \theta. \quad (19)$$

B. An exact first integral

Dividing Eq. (19) by Eq. (18):

$$\frac{\cos \theta d\theta}{\sin \theta} = -\beta \frac{dT}{T} \implies \boxed{T(s)^{\beta} \sin \theta(s) = \text{const.}} \quad (20)$$

Relation (20) is exact within RFT, with no small-angle approximation. It states that the thread is most tilted where the tension is lowest and straightens where the tension grows. The coordinates follow by quadrature:

$$x(s) = \int_0^s \sin \theta ds' = \frac{1}{\eta_{\parallel} U} \int_{T_0}^{T(s)} \tan \theta dT, \quad z(s) = \int_0^s \cos \theta ds'. \quad (21)$$

[TO DO: Alex and Vlad to check Eq. (20) on steady DER simulations: $T^{\beta} \sin \theta$ must be constant along the thread. This is the cleanest test of the code.]

C. Boundary conditions for N threads

Assume a symmetric configuration: the tips of the N threads lie on a horizontal circle of radius R , at azimuths $2\pi j/N$. The radial Coulomb force on one tip due to the other $N - 1$ tips is

$$F_C = \frac{k_e q^2}{4R^2} S_N, \quad S_N \equiv \sum_{j=1}^{N-1} \frac{1}{\sin(\pi j/N)}, \quad (22)$$

with $S_2 = 1$, $S_4 = 1 + 2\sqrt{2} \approx 3.83$, and $S_8 \approx 11.2$.

To derive Eq. (22), place the tip under consideration at azimuth 0, $\mathbf{x}_0 = R(1, 0)$, and tip j at $\varphi_j = 2\pi j/N$, $\mathbf{x}_j = R(\cos \varphi_j, \sin \varphi_j)$, with $j = 1, \dots, N - 1$. The separation vector and its length are

$$\mathbf{x}_0 - \mathbf{x}_j = R(1 - \cos \varphi_j, -\sin \varphi_j), \quad d_j = \|\mathbf{x}_0 - \mathbf{x}_j\| = R\sqrt{2(1 - \cos \varphi_j)} = 2R \sin(\pi j/N). \quad (23)$$

The radial component of the Coulomb force exerted by tip j is

$$F_{C,j} = \frac{k_e q^2}{d_j^3} R(1 - \cos \varphi_j) = \frac{k_e q^2 2R \sin^2(\pi j/N)}{8R^3 \sin^3(\pi j/N)} = \frac{k_e q^2}{4R^2} \frac{1}{\sin(\pi j/N)}. \quad (24)$$

The tangential components, $\propto \sin \varphi_j$, cancel in pairs because tips j and $N - j$ are symmetric about the radius through the tip considered. Summing $F_{C,j}$ over j gives Eq. (22). **[TO DO: Alex and Vlad compute S_8 exactly.]**

At the tip, the tension balances the electrostatic and Coulomb forces:

$$T_L \cos \theta_L = qE, \quad T_L \sin \theta_L = F_C. \quad (25)$$

At the spider, $NT_0 \cos \theta_0 = mg + \zeta_s U$.

D. Small-angle limit

For $\theta \ll 1$: $\cos \theta \approx 1$, hence $T(s) = T_0 + \eta_{\parallel} U s$ and $T_L - T_0 = \eta_{\parallel} U L$, and Eq. (8) is recovered as a consistency condition. We define the tension ratio

$$\tau \equiv \frac{T_0}{T_L} = \frac{mg + \zeta_s U}{NqE}. \quad (26)$$

From Eq. (20), $\theta(s) = \theta_L [T_L/T(s)]^\beta$, and the tip radius follows by integration:

$$R = \int_0^L \theta ds = \theta_L L g(\tau), \quad g(\tau) \equiv \frac{\tau^{1-\beta} - 1}{(\beta - 1)(1 - \tau)}, \quad (27)$$

with $g \rightarrow 1$ as $\tau \rightarrow 1$ (straight, uniformly tilted thread). With $\theta_L \approx F_C/(qE)$ and Eq. (22) we obtain **the opening of the bundle**:

$$\boxed{\frac{R}{L} = \left[\frac{S_N \Lambda g(\tau)}{4} \right]^{1/3}, \quad \theta_L = \frac{S_N \Lambda}{4} \left(\frac{L}{R} \right)^2} \quad (28)$$

The solution is valid for $R \ll L$.

a. Order of magnitude. For $N = 2$, $q = 0.6$ nC, $E = 7.41 \times 10^3$ V/m, $L = 0.5$ m, and $\tau \approx 1$: $\Lambda \approx 2.9 \times 10^{-3}$, $R/L \approx 0.09$, $R \approx 4.5$ cm, and $\theta_L \approx 0.09$ rad. The small-angle approximation is therefore justified.

b. Testable predictions.

1. $R \propto q^{1/3} E^{-1/3} L^{1/3}$ at fixed τ , a power law to be checked on a log-log plot.
2. The dependence on N enters only through $S_N^{1/3}$ and through τ . For large N , $S_N \simeq (2N/\pi) [\ln(2N/\pi) + \gamma]$, with γ the Euler–Mascheroni constant. This follows from $1/\sin(\pi j/N) \approx N/(\pi j)$ for the dominant near neighbours and the Euler–Maclaurin formula; the relative error is 0.6% for $N = 4$, 0.1% for $N = 8$, and 0.02% for $N = 16$. **[TO DO: Alex and Vlad: tabulate S_N for $N = 2, \dots, 16$.]**
3. A uniform flow does not change R (Sec. III); turbulence does, and the statistics of R are a numerical result.

The assumption that all tips lie on the same circle is reasonable for $N \leq 8$ and tip charge. For uniform charge, Eq. (7) acquires the distributed term $\mathbf{f}_\tau$ and no longer has a simple closed-form solution; that case remains numerical in the present work.

V. NUMERICAL METHOD

A. Discretization

The simulations use the discrete elastic rods (DER) method [5] as implemented in [1]. The spider is node $\mathbf{x}_0$; each thread j is discretized into N_t edges with nodes $\mathbf{x}_{(j-1)N_t+1}, \dots, \mathbf{x}_{jN_t}$, so the system has $NN_t + 1$ nodes. Each edge carries a twist angle θ^k . The degrees of freedom are

$$\boldsymbol{\xi} = [\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_{NN_t}, \theta^0, \dots, \theta^{NN_t-1}]^T \in \mathbb{R}^{n_{\text{dof}}}, \quad n_{\text{dof}} = 3(NN_t + 1) + NN_t. \quad (29)$$

The elastic energy $E_{\text{el}}(\boldsymbol{\xi}) = E_s + E_b + E_t$ contains stretching, bending, and twisting contributions [5]. With edge vectors $\mathbf{e}^k$, reference edge length $l_0 = L/N_t$, Voronoi length Δl_i , cross-section $A = \pi r^2$, and moments $I = \pi r^4/4$, $J = \pi r^4/2$,

$$E_s = \frac{1}{2} Y A \sum_k \left(\frac{\|\mathbf{e}^k\|}{l_0} - 1 \right)^2 l_0, \quad (30)$$

$$E_b = \sum_i \frac{YI}{\Delta l_i} \|\boldsymbol{\kappa}_i - \boldsymbol{\kappa}_i^0\|^2, \quad (\boldsymbol{\kappa}\mathbf{b})_i = \frac{2\mathbf{e}^{i-1} \times \mathbf{e}^i}{\|\mathbf{e}^{i-1}\| \|\mathbf{e}^i\| + \mathbf{e}^{i-1} \cdot \mathbf{e}^i}, \quad (31)$$

$$E_t = \sum_i \frac{GJ}{\Delta l_i} m_i^2, \quad m_i = \theta^i - \theta^{i-1} + \Delta m_{i,\text{ref}}, \quad (32)$$

where $\boldsymbol{\kappa}_i$ collects the projections of the curvature binormal $(\boldsymbol{\kappa}\mathbf{b})_i$ on the material frames of the two adjacent edges, $\boldsymbol{\kappa}_i^0 = \mathbf{0}$ for naturally straight threads, $G = Y/[2(1 + \nu)]$, and $\Delta m_{i,\text{ref}}$ is the reference twist of the parallel-transported frames. The bending and twisting stencils do not connect different threads through the spider node, which is a free junction. The stretching stiffness is chosen large to enforce inextensibility. The external force vector is

$$\mathbf{f}^{\text{ext}}(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}) = \mathbf{W} + \mathbf{F}_v + \mathbf{F}_r + \mathbf{F}_l, \quad (33)$$

with, at node k with Voronoi length Δl_k and node tangent $\mathbf{t}_k$:

$$\mathbf{F}_{v,k} = -[\eta_{\parallel} \mathbf{t}_k \mathbf{t}_k^T + \eta_{\perp} (\mathbb{I} - \mathbf{t}_k \mathbf{t}_k^T)] \Delta l_k (\mathbf{v}_k - \mathbf{u}(\mathbf{x}_k, t)), \quad (34)$$

$$\mathbf{F}_{l,k} = q_k E(z_k) \hat{\mathbf{z}}, \quad (35)$$

the Coulomb term from Eq. (6), and the Stokes drag at the spider node. As in [1], the charge of the spider does not act on the thread nodes (sum over $i \neq 0$); the spider still feels the thread charges. Including it changes the forces by $\lesssim Q_s/q \sim 10^{-3}$. The nodal charges are $q_k = q$ at the tip nodes (case a) or $q_k = q \Delta l_k/L$ (case b), and $q_0 = Q_s$.

B. Time integration

The equations of motion $m_i \ddot{\xi}_i + \partial E_{\text{el}}/\partial \xi_i - \mathbf{f}_i^{\text{ext}} = 0$ are integrated with the implicit (backward) Euler method. At each step we solve the nonlinear residual

$$\mathbf{F}(\boldsymbol{\xi}^{n+1}) \equiv \frac{\mathbf{M}}{\Delta t} \left(\frac{\boldsymbol{\xi}^{n+1} - \boldsymbol{\xi}^n}{\Delta t} - \dot{\boldsymbol{\xi}}^n \right) + \nabla E_{\text{el}}(\boldsymbol{\xi}^{n+1}) - \mathbf{f}^{\text{ext}}(\boldsymbol{\xi}^{n+1}) = \mathbf{0} \quad (36)$$

by Newton–Raphson, with Jacobian

$$\mathbf{J} = \frac{\mathbf{M}}{\Delta t^2} + \nabla^2 E_{\text{el}} - \frac{\partial \mathbf{f}^{\text{ext}}}{\partial \boldsymbol{\xi}}. \quad (37)$$

The drag term is linear in the velocity $(\boldsymbol{\xi}^{n+1} - \boldsymbol{\xi}^n)/\Delta t$, so its contribution to $\mathbf{J}$ is $\mathbf{C}/\Delta t$, with $\mathbf{C}$ the block-diagonal RFT resistance matrix (the dependence of $\mathbf{t}_k$ on $\boldsymbol{\xi}$ may be lagged). The Coulomb Jacobian is dense for uniform charge and is computed analytically from Eq. (6). The step size is adaptive, as in [1]: divided by 10 on Newton failure, multiplied by 10 after 10 successful steps, and capped at Δt_{max} . The rotational inertia of the twist degrees of freedom is $\rho_t J l_0/2$ per edge, with Poisson ratio $\nu = 0.5$. A run is aborted if Δt falls below 10^{-9} s. The default numerical parameters are listed in Table XII. For the isotropic, naturally straight threads considered here, the twist angles stay

below 10^{-8} rad, as expected for free ends. The procedure is summarized in Algorithm 1; Algorithm 2 describes the parameter sweep used for the phase diagram.

Algorithm 1 Implicit DER time stepping for a multi-thread ballooning spider in an air flow

Require: parameters $\mathcal{P} = \{N, N_t, L, r, m, r_s, q, \text{charge model}, Y, \mu\}$, field $E(z)$, air velocity $\mathbf{u}(\mathbf{x}, t)$
Require: initial state $\boldsymbol{\xi}^0$ (threads vertical, spaced $d_0 = 100 \mu\text{m}$), $\boldsymbol{\xi}^0 = \mathbf{0}$
Require: $\Delta t_0, \Delta t_{\max}$, tolerance ϵ , max. Newton iterations K , final time t_{end}
Ensure: trajectory $\{\boldsymbol{\xi}^n, t^n\}$, steady-state flag

- 1: $n \leftarrow 0, t \leftarrow 0, \Delta t \leftarrow \Delta t_0, c \leftarrow 0$ ▷ c : consecutive successful steps
- 2: compute lumped mass matrix $\mathbf{M}$ (spider mass at node 0, thread mass on other nodes)
- 3: **while** $t < t_{\text{end}}$ **do**
- 4: $\boldsymbol{\xi}^{(0)} \leftarrow \boldsymbol{\xi}^n; k \leftarrow 0$ ▷ Newton initial guess
- 5: **repeat**
- 6: update reference frames by parallel transport; compute material frames from $\boldsymbol{\xi}^{(k)}$
- 7: $\mathbf{v}^{(k)} \leftarrow (\boldsymbol{\xi}^{(k)} - \boldsymbol{\xi}^n)/\Delta t$
- 8: $\mathbf{f}^{\text{ext}} \leftarrow \mathbf{W} + \mathbf{F}_v(\boldsymbol{\xi}^{(k)}, \mathbf{v}^{(k)}, \mathbf{u}) + \mathbf{F}_r(\boldsymbol{\xi}^{(k)}) + \mathbf{F}_l(\boldsymbol{\xi}^{(k)})$ ▷ Eqs. (6), (34)
- 9: assemble residual $\mathbf{F}$ (Eq. (36)) and Jacobian $\mathbf{J}$
- 10: solve $\mathbf{J} \Delta \boldsymbol{\xi} = \mathbf{F}; \boldsymbol{\xi}^{(k+1)} \leftarrow \boldsymbol{\xi}^{(k)} - \Delta \boldsymbol{\xi}; k \leftarrow k + 1$
- 11: **until** $\|\mathbf{F}\|_1 < \epsilon$ **or** $k = K$
- 12: **if** $\|\mathbf{F}\|_1 \geq \epsilon$ **then** ▷ Newton failed
- 13: $\Delta t \leftarrow \Delta t/10; c \leftarrow 0$ **and retry the step**
- 14: **else**
- 15: $\boldsymbol{\xi}^{n+1} \leftarrow \boldsymbol{\xi}^{(k)}; \dot{\boldsymbol{\xi}}^{n+1} \leftarrow (\boldsymbol{\xi}^{n+1} - \boldsymbol{\xi}^n)/\Delta t; t \leftarrow t + \Delta t; n \leftarrow n + 1$
- 16: $c \leftarrow c + 1$; **if** $c = 10$ **then** $\Delta t \leftarrow \min(10 \Delta t, \Delta t_{\max}), c \leftarrow 0$
- 17: **save** $(t, \boldsymbol{\xi}^n)$ at fixed output interval ▷ HDF5, for the ML part
- 18: **if** $|\dot{z}_0^n - \dot{z}_0^{n-1}|/|\dot{z}_0^n| < \delta$ over a window t_w **then** ▷ steady state
- 19: **return** trajectory, steady-state flag = true
- 20: **end if**
- 21: **end if**
- 22: **end while**
- 23: **return** trajectory, steady-state flag = false

Algorithm 2 Ensemble estimate of the take-off probability

Require: grids $\{N\}, \{\bar{F}_l\}$ (via q), $\{\sigma_w\}, \{\ell\}$; release height z_0 , target height h ; M realizations per point

- 1: **for** each $(N, \bar{F}_l, \sigma_w, \ell)$ **do**
- 2: $n_{\text{up}} \leftarrow 0$
- 3: **for** $i = 1, \dots, M$ **do**
- 4: draw seed s_i ; build the modes $\{\mathbf{k}_n, \mathbf{a}_n, \mathbf{b}_n, \omega_n\}$ of Eq. (14)
- 5: hold the spider node fixed at z_0 and relax the bundle in still air (Algorithm 1 with $\boldsymbol{\xi}_0$ clamped, to steady state); release at $t = 0$ with $\mathbf{u}(\mathbf{x}, t)$ switched on
- 6: run Algorithm 1 with $\mathbf{u}(\mathbf{x}, t)$ from Eq. (14), stopping when $z_0(t) \geq h, z_0(t) \leq 0$, or $t \geq t_{\text{end}}$
- 7: **if** $z_0 \geq h$ **then** $n_{\text{up}} \leftarrow n_{\text{up}} + 1$
- 8: record: outcome, exit time, time series of $R(t), \theta_L(t), d_{\min}(t)$; flag ENTANGLED if $d_{\min} < 2r$
- 9: **end for**
- 10: $\hat{P} \leftarrow n_{\text{up}}/M$ with Wilson 95% interval; compare with Eq. (17)
- 11: **end for**

Algorithm 1 is the one of [1], with three explicit additions: the air velocity $\mathbf{u}(\mathbf{x}, t)$ in the drag, evaluated at every node, the stopping tests, and systematic data saving. The realizations of Algorithm 2 are independent and run in parallel. The open-source code DisMech [6] implements the DER core. If adapting it takes more than a few days, a bead-spring chain with stiff springs, RFT forces on nodes, and the same implicit scheme is sufficient, since bending is negligible.

VI. LEARNED SURROGATE MODEL

This section belongs to the ML part (Octavian). Only the structure and the minimum content expected by a physics audience are fixed here.

A. Data set

The training data are the trajectories produced by Algorithm 2, stored in the format of Table XI. **[TO DO: Number of runs, parameter ranges, time step between saved frames, total number of frames.]**

TABLE IV. Data splits. Generalization is tested on numbers of threads and parameter values not seen during training.

Split	N	q (nC)	σ_w (m/s)	Runs	Frames
Train	?	?	?	?	?
Validation	?	?	?	?	?
Test (interpolation)	?	?	?	?	?
Test (unseen N)	?	?	?	?	?
Test (unseen σ_w)	?	?	?	?	?

B. Graph representation

Each node of the discretized bundle is a graph node, with features **[TO DO: position relative to the spider, velocity history, charge q_k , node type (spider, interior, tip), and global parameters E , σ_w , ℓ ; the air velocity $u(x_k, t)$ at each node as input]**. Two types of edges are used: (i) *material edges* along each thread, and (ii) *interaction edges* that carry the Coulomb and hydrodynamic coupling between threads. **[TO DO: Radius graph versus Ewald/long-range scheme; justify with the $1/r^2$ decay of Eq. (6).]**

C. Architecture and training

[TO DO: Encoder–processor–decoder architecture [14] with long-range message passing [13]; number of layers, latent size; prediction target (acceleration, velocity, or next position); loss; noise injection; optimizer and schedule; hardware and training time.]

D. Evaluation metrics

- one-step and rollout mean squared error on node positions;
- error in the steady-state observables V , R/L , and θ_L ;
- violation of the analytical constraints: deviation of $T^\beta \sin \theta$ from a constant [Eq. (20)], of R/L from Eq. (28), and of the rise/fall classification from Eq. (13);
- computational cost: wall-clock time per run and per estimate of P , one-time costs of data generation and training, break-even number of evaluations (40), and error of P at equal compute budget (Sec. VII C 1).

VII. RESULTS

[TO DO: Write a short paragraph introducing the three parts of the results: validation, comparison with the analytical predictions, and the learned surrogate.]

[TO DO: Final figures: regenerate all figures for publication once the data are final. The current ones are working plots. Common style for all figures (PRE column width, consistent fonts and line widths,

colour-blind-safe palette, larger markers and labels); readable axes (no offset exponents, shared limits where panels are compared); insets or residual panels where the agreement with theory is the message (Figs. 2, 3); a proper 3D rendering of the bundles (Fig. 4); vector PDF output.]

A. Validation of the numerical model

1. Analytical check: terminal velocity

From Eq. (8), for a vertical thread in still air and $\zeta_s \ll N\eta_{\parallel}L$:

$$U \approx \frac{NqE - mg}{N\eta_{\parallel}L} = \frac{\ln(L/r) - \frac{1}{2}}{2\pi} \frac{NqE - mg}{N\mu L} \approx 2.2 \frac{NqE - mg}{N\mu L}. \quad (38)$$

Habchi and Jawed normalize the velocity by $v_{\text{charac}} = (NqE - mg)/(\mu NL)$ and obtain numerically $\bar{v}_t \approx 2.2$ for one thread [1]. Equation (38) **explains this constant analytically**: it is $[\ln(L/r) - 1/2]/(2\pi)$. Keeping the Stokes drag of the body, the exact value for straight vertical threads is

$$\bar{v}_t^{\text{st}} = \frac{\mu NL}{N\eta_{\parallel}L + \zeta_s}, \quad (39)$$

which reduces to Eq. (38) for $\zeta_s \ll N\eta_{\parallel}L$. The correction is not small for short threads: $\zeta_s/(\eta_{\parallel}L) = 0.37$ for $L = 0.1$ m and 0.08 for $L = 0.5$ m. For $N > 1$ the threads tilt, the normal component $\eta_{\perp} > \eta_{\parallel}$ enters the drag, and $\bar{v}_t$ falls below Eq. (39).

Table V compares both expressions with the simulations. The numerical values agree with Eq. (39) within 0.36–0.45%, consistent with the discretization error at $N_t = 100$ (Table VII); Eq. (38) overestimates $\bar{v}_t$ by up to 36% for $L = 0.1$ m, because it neglects the body drag.

TABLE V. Dimensionless terminal velocity $\bar{v}_t = U\mu NL/(NqE - mg)$ for a single vertical thread ($N = 1$, $r = 300$ nm, $m = 1$ mg, $Q_s = 0$, tip charge with $\bar{F}_l = 2$, still air, constant field $E = 7.41 \times 10^3$ V/m, $N_t = 100$). Analytical values from Eq. (38) (thread drag only) and Eq. (39) (with the Stokes drag of the body); numerical values from Algorithm 1 at steady state. The relative error is with respect to Eq. (39).

L (m)	$\ln(L/r)$	$\bar{v}_t$, Eq. (38)	$\bar{v}_t$, Eq. (39)	$\bar{v}_t$ numerical	Relative error (%)
0.1	12.72	1.944	1.423	1.428	0.36
0.5	14.33	2.201	2.032	2.041	0.45
1.0	15.02	2.311	2.214	2.224	0.45

2. Reproduction of the validation in [1]

- Test 1.** One thread of 0.5 m in the chamber field of Eq. (2), with the mass of the spiders of Morley and Gorham, $m = 0.9$ mg [3], and $Q_s = 0$. Habchi and Jawed report a steady velocity of ≈ 8.5 cm/s reached after ≈ 0.1 s [1], but do not state the charge or its distribution for this case. With tip charge, the tip sits at $z \approx L$, where $E \simeq E_{\infty} = 7.41 \times 10^3$ V/m, so the steady velocity follows from Eq. (8); we choose $q = (mg + U\zeta_{\text{tot}})/E_{\infty} = 1.2433$ nC, with $\zeta_{\text{tot}} = \eta_{\parallel}L + \zeta_s$, to target $U = 8.5$ cm/s. This test therefore checks the force balance and the transient, not the value 8.5 cm/s independently. We run two variants: (a) the charge at the tip, and (b) the same charge on the spider node, which mimics the one-dimensional model of [3] and places the charge in the enhanced field near the prominence.
- Test 2.** One point of Fig. 6 in [1]: $N = 2$, tip charge, $\bar{F}_l = 2$ ($q = 1.324$ nC), $m = 1$ mg, $Q_s = 3$ pC, constant field $E = 7.41 \times 10^3$ V/m. Target: $\bar{v}_t \approx 2$, read from the figure.
- Test 3.** Discretization convergence for test 2: $N_t = 25, 50, 100, 200$ (Table VII).
- Code checks.** Finite-difference verification of all gradients, Hessians, and Jacobians; terminal velocity against Eq. (8), with and without Q_s ; the first integral (20); and the uniform-flow invariance of Sec. III.

For the uniform-flow check, both runs use a fixed step $\Delta t = 3 \times 10^{-5}$ s and the default Newton tolerance, and the shapes are compared in the spider frame every 0.1 s up to $t = 2$ s. With the same initial state relative to the air, the shapes coincide to $8.5 \times 10^{-12} L$, the level of round-off. With both runs started at rest in the ground frame, the difference decays exponentially with time constant 0.24 s, four times the translational time $t_s = m/(N\eta_{\parallel}L + \zeta_s) = 0.059$ s; the difference in the tip radius $R(t)$ alone decays with the same time constant (0.238 s, against 0.236 s for the full shape), which identifies the slowest mode as the lateral opening of the bundle, driven by the Coulomb force against the normal drag. **[TO DO: Analytical estimate of this time constant.]**

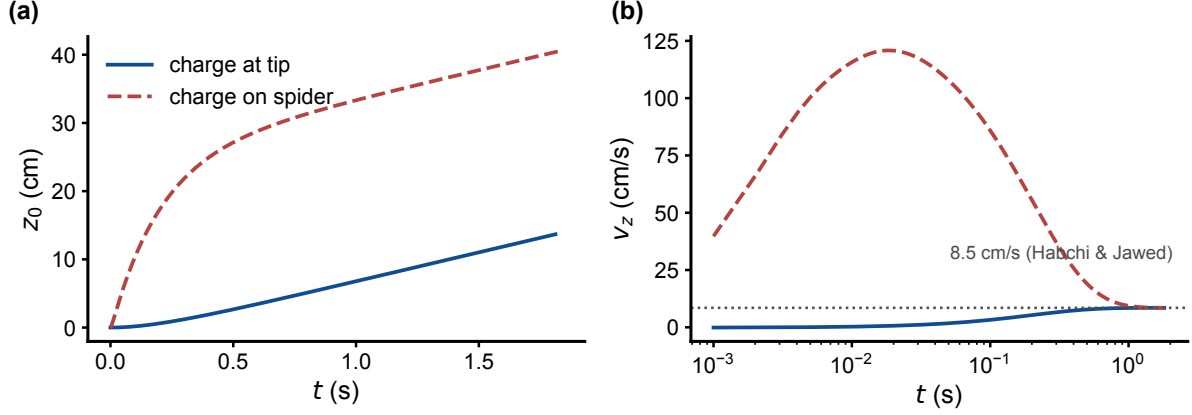


FIG. 1. Test 1: take-off of a spider ($m = 0.9$ mg, $Q_s = 0$) with one thread ($L = 0.5$ m, $N_t = 100$) in the chamber field of Eq. (2), with $q = 1.2433$ nC. (a) Altitude $z_0(t)$ and (b) vertical velocity $v_z(t)$ of the spider node, for the charge at the thread tip (solid) and on the spider node (dashed). The dotted line marks the steady velocity ≈ 8.5 cm/s reported in [1]. With tip charge the velocity relaxes on the time scale $t_s = m/(\eta_{\parallel}L + \zeta_s) = 0.20$ s; with the charge on the spider, the enhanced field near the prominence ejects the spider at up to 1.21 m/s before it settles.

TABLE VI. Reproduction of the validation cases of [1].

Test	Configuration	Reference value	This work	Error (%)
1a	$N = 1$, $m = 0.9$ mg, tip charge, Eq. (2)	$U \approx 8.5$ cm/s [1]	8.502 cm/s	0.02
1a	same	$t_{95} = 3t_s = 0.597$ s (theory)	0.604 s	1.1
1a	same	time to steady state ≈ 0.1 s [1]	0.604 s (t_{95})	—
1b	same, charge on the spider node	$U \approx 8.5$ cm/s [1]	8.516 cm/s	0.19
1b	same	—	peak 1.21 m/s; within 5% of U after 1.14 s	—
2	$N = 2$, tip charge, $\bar{F}_l = 2$, $N_t = 100$	$\bar{v}_t \approx 2$ (read from Fig. 6 of [1])	2.088	4.4
check	$N = 1$, constant E , $Q_s = 0$	Eq. (8): 9.513 cm/s	9.520 cm/s	0.076
check	$N = 1$, constant E , $Q_s = 3$ pC	Eq. (8) with $Q_s E$: 10.044 cm/s	10.050 cm/s	0.066
check	$N = 2$, first integral (20)	$T^\beta \sin \theta = \text{const.}$	spread 0.019%	—
check	$N = 4$, uniform flow $w = 0.5$ m/s, same initial state relative to the air	$\Delta V = 0.5$ m/s; same shape at all t	$\Delta V = 0.50000000005$ m/s; max. shape deviation $8.5 \times 10^{-12} L$ over $t \leq 2$ s	10^{-8}
check	same, both runs started at rest in the ground frame	transient, same steady state	deviation $2.3 \times 10^{-3} L$ at $t = 0.1$ s, $9 \times 10^{-7} L$ at $t = 2$ s; decay time 0.24 s	—
check	gradients, Hessians, Jacobians	finite differences	agreement to $\sim 10^{-10}$	—

TABLE VII. Discretization convergence for test 2 ($N = 2$, tip charge, $\bar{F}_l = 2$, $Q_s = 3$ pC). Δ is the relative difference with respect to the finest mesh, $N_t = 200$. The differences halve with each refinement (first-order convergence); Richardson extrapolation gives $\bar{v}_t \rightarrow 2.078$. CPU time: one core of an Apple M4 Pro.

N_t (edges/thread)	$\bar{v}_t$	$\Delta\bar{v}_t$ (%)	R/L	$\Delta(R/L)$ (%)	CPU time (s)
25	2.1196	1.75	0.14287	0.48	5.6
50	2.0986	0.74	0.14248	0.20	11.5
100	2.0883	0.25	0.14229	0.067	25.7
200	2.0832	—	0.14220	—	51.6

B. Analytical vs. numerical comparison

[TO DO: Produce Figs. 2–8 and Table VIII. Figures 2 and 3 use single still-air runs of Algorithm 1; Figs. 5–7 use Algorithm 2.]

Table VIII compares the steady opening of tip-charged bundles with Eq. (28). The simulated tip radius is 0.8%, 1.3%, and 1.8% below the prediction for $N = 2, 4$, and 8, and the tip angle θ_L is 1.8–3.7% above it, although R/L reaches 0.20. The deviations grow with R/L , as expected from the small-angle approximation. The normalized velocity is 1.4–3.3% below the straight-thread value of Eq. (39), which measures the extra drag due to the tilt of the threads. The threads never come closer than 4 mm, so no entanglement occurs in still air. Figure 2 extends the comparison to $\bar{F}_l = 1.2$ –5: the data for all N collapse on the line of slope 1 up to $R/L \approx 0.2$, and fall below it at larger openings, by about 10% at $R/L \approx 0.3$, where the small-angle approximation breaks down.

TABLE VIII. Steady-state observables for tip-charged bundles in still air ($L = 0.5$ m, $m = 1$ mg, $Q_s = 0$, $\bar{F}_l = 2$, constant $E = 7.41 \times 10^3$ V/m, $N_t = 100$). τ uses the simulated U . Λ , τ , and $(R/L)_{\text{an}}$ from Eqs. (27)–(28); the remaining columns from simulation. d_{min} : minimum distance between threads.

N	q (nC)	Λ	τ	$(R/L)_{\text{an}}$	$(R/L)_{\text{num}}$	θ_L (rad)	$\bar{v}_t$	d_{min} (μm)
2	1.324	6.42×10^{-3}	0.520	0.1433	0.1422	0.0796	2.084	7497
4	0.662	3.21×10^{-3}	0.510	0.1790	0.1766	0.0987	2.103	6584
8	0.331	1.61×10^{-3}	0.505	0.2039	0.2003	0.1123	2.106	4021

C. Learned surrogate

[TO DO: Octavian: results text.]

TABLE IX. Surrogate accuracy on the test splits of Table IV. Rollout errors after ? steps. Baselines: ? (e.g. local-only GNN without long-range edges).

Model	Split	Rollout MSE	$\Delta V/V$ (%)	$\Delta(R/L)$ (%)	$ \Delta P $	Speed-up
Local GNN	interp.	?	?	?	?	?
Local GNN	unseen N	?	?	?	?	?
Long-range GNN	interp.	?	?	?	?	?
Long-range GNN	unseen N	?	?	?	?	?
Local GNN	unseen σ_w	?	?	?	?	?
Long-range GNN	unseen σ_w	?	?	?	?	?

1. Computational cost

a. Protocol. The surrogate and the simulator are compared on the same tasks: (i) one complete run, from release until the spider reaches h or the ground, and (ii) one estimate of the take-off probability P from M realizations. Wall-

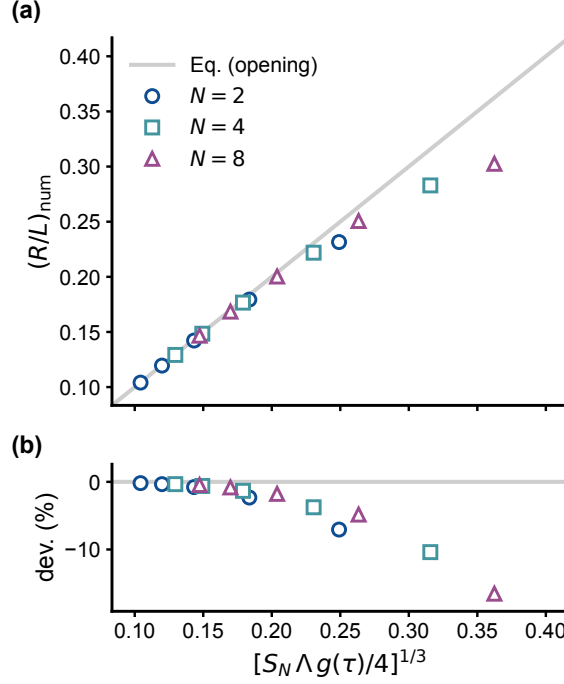


FIG. 2. Opening of the bundle. (a) Tip radius from simulation, $(R/L)_{\text{num}}$, versus the small-angle prediction of Eq. (28), for $N = 2$ (circles), 4 (squares), and 8 (triangles), and $\bar{F}_l = 1.2, 1.5, 2, 3, 5$ (tip charge, still air, $m = 1$ mg, $Q_s = 0$, constant $E = 7.41 \times 10^3$ V/m). The solid line has slope 1. (b) Relative deviation from the prediction: below 1% for small openings, growing to about -17% at the largest opening ($N = 8$, $\bar{F}_l = 5$), which marks the breakdown of the small-angle approximation.

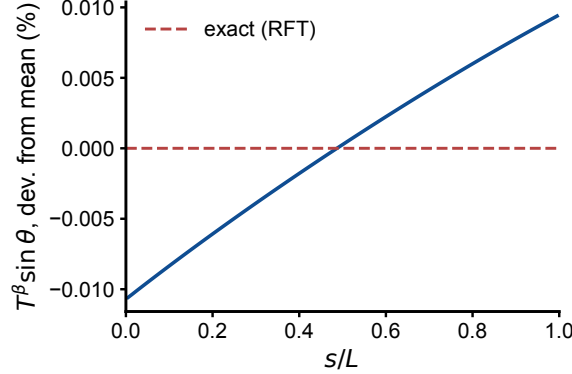


FIG. 3. Numerical check of the first integral of Eq. (20). The quantity $T^\beta \sin \theta$, normalized by its mean value, along one thread at steady state ($N = 2$, tip charge, $q = 1.324$ nC, $\bar{F}_l = 2$, $Q_s = 0$), for $N_t = 100$. Shown is the deviation from the mean in percent; the exact RFT result is zero (dashed line). The spread over the interior edges is 0.019%.

clock times are reported for both methods in their optimized setting: the simulator with independent realizations run in parallel on all CPU cores, the surrogate with realizations batched on the GPU. Hardware: ? (CPU, cores, GPU, memory).

b. Total cost and break-even. Let c_{sim} and c_{sur} be the costs of one estimate of P with the simulator and with the surrogate, and C_{data} and C_{train} the one-time costs of generating the training data and training the network. The surrogate is cheaper overall after

$$Q^* = \frac{C_{\text{data}} + C_{\text{train}}}{c_{\text{sim}} - c_{\text{sur}}} \quad (40)$$

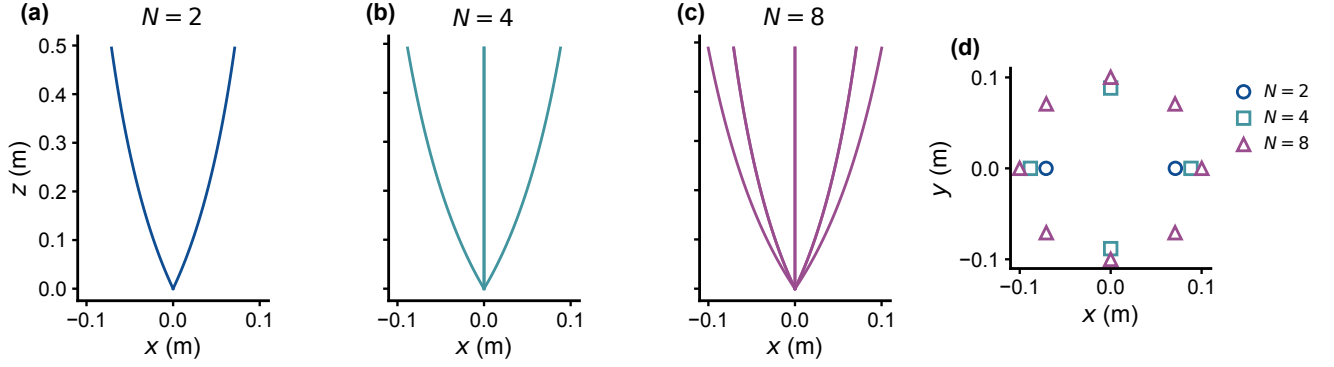


FIG. 4. Steady shapes of tip-charged bundles in still air ($L = 0.5$ m, $m = 1$ mg, $Q_s = 0$, $\bar{F}_l = 2$, constant $E = 7.41 \times 10^3$ V/m, $N_t = 100$; parameters of Table VIII). (a)–(c) Side view of the threads in the frame of the spider for $N = 2, 4$, and 8 . (d) Top view of the tips, which lie on a circle of radius R , as assumed in Sec. IV.

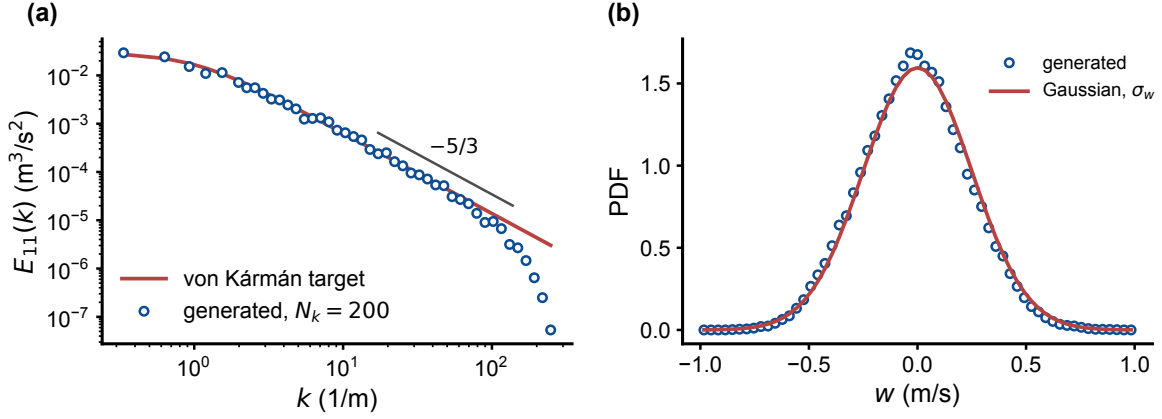


FIG. 5. Validation of the synthetic turbulence of Eq. (14), for $\sigma_w = 0.25$ m/s, $\ell = 1$ m, and $N_k = 200$ modes (average over 50 realizations). (a) Longitudinal one-dimensional spectrum of w along a vertical line versus the spectrum implied by Eq. (15); the generated spectrum is shown up to the resolution cut-off $k_{\max} = 2\pi/(5\Delta l)$ and falls below the target in the last decade because of the finite shell width near $k_{\max}$. (b) Probability density of w with the Gaussian of standard deviation σ_w . **[TO DO: Regenerate with the renormalized amplitudes of Stage B.]**

evaluations of P . Q^* must be compared with the number of evaluations actually needed, for example the number of points in Fig. 6, or in a finer sweep or an inverse problem.

c. Accuracy at equal cost. The reference value P_{ref} is estimated with the simulator using $M_{\text{ref}} = ?$ realizations. For a given compute budget B , the simulator affords $M_{\text{sim}}(B)$ realizations and the surrogate $M_{\text{sur}}(B) \gg M_{\text{sim}}(B)$. The error of the surrogate estimate contains a statistical part, $\propto M_{\text{sur}}^{-1/2}$, and a systematic part (model bias) that does not decrease with M . The surrogate is useful only where its bias is smaller than the statistical error of the simulator at the same budget.

TABLE X. Computational cost of the simulator (Algorithm 1) and of the surrogate, for $N = ?$, $\sigma_w = ?$ m/s. $|\Delta P|$: mean absolute error of P with respect to P_{ref} , over the points of Fig. 6, at a fixed budget $B = ?$ s per point. Hardware as stated in the text.

Method	Hardware	Time per run (s)	Time per P ($M = ?$) (s)	C_{data} (h)	C_{train} (h)	Q^*	$ \Delta P $ at budget B
Simulator	CPU, ? cores	?	?	—	—	—	?
Surrogate	GPU, ?	?	?	?	?	?	?

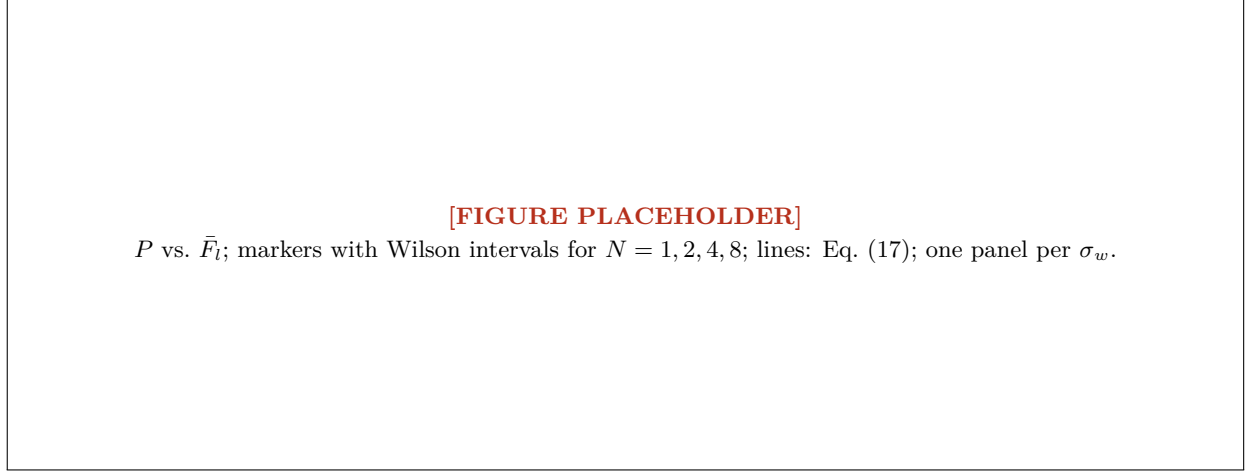


FIG. 6. Take-off probability versus lift-to-weight ratio, for (a) $\sigma_w = ?$, (b) $?$, and (c) $?$ m/s ($\ell = ?$ m, $z_0 = ?$ m, $h = ?$ m, tip charge, $m = 1$ mg). Markers: ensemble estimates from Algorithm 2 ($M = ?$ realizations) with 95% Wilson intervals, for $N = 1$ (circles), 2 (squares), 4 (triangles), and 8 (diamonds). Lines: drift-diffusion reference, Eq. (17). The vertical dashed line marks the deterministic threshold $\bar{F}_l = 1$.

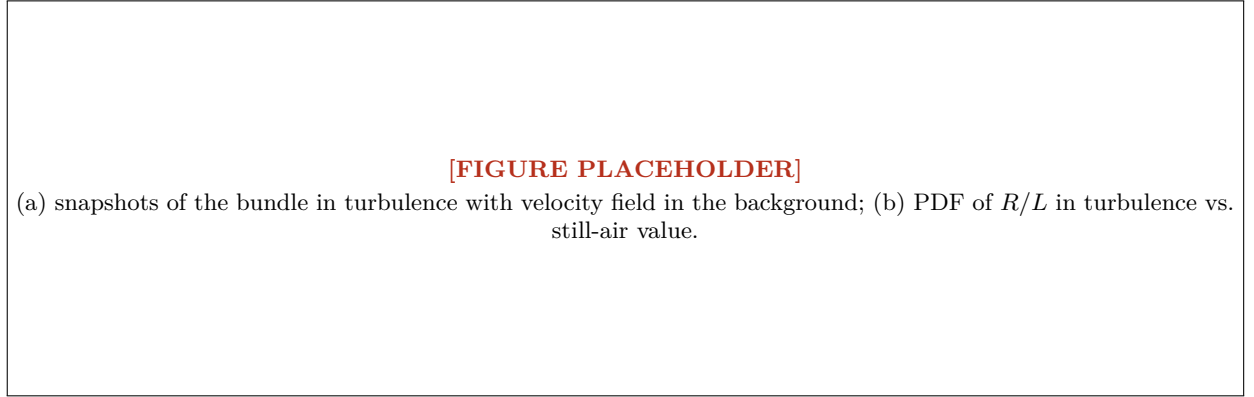


FIG. 7. Deformation of the bundle by turbulence ($N = 4$, $q = ?$ nC, $\sigma_w = ?$ m/s). (a) Snapshots at $?$ s intervals, with the vertical velocity in a plane through the spider. (b) Probability density of the tip radius R/L , compared with the still-air value of Eq. (28).

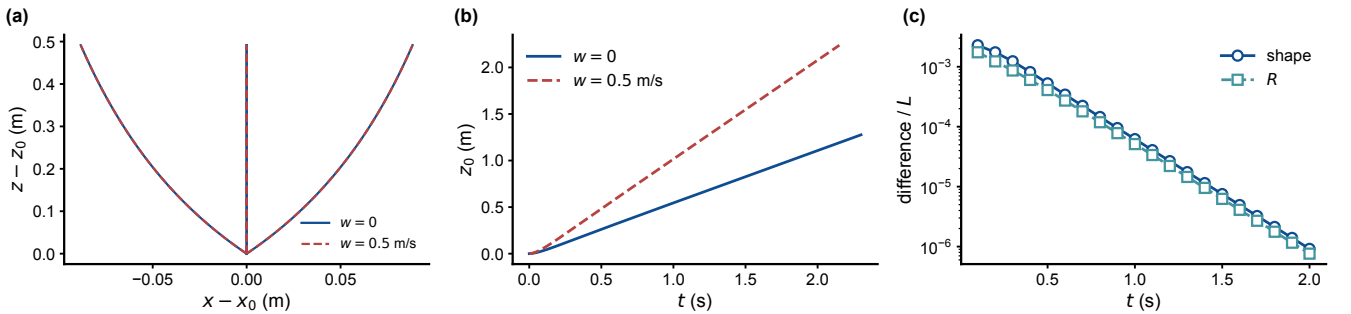


FIG. 8. Code test: shape invariance under a uniform updraft ($N = 4$, tip charge, $q = 0.662$ nC, $\bar{F}_l = 2$, $Q_s = 0$, constant $E = 7.41 \times 10^3$ V/m, $N_t = 100$). (a) Steady-state configurations in the frame of the spider for $w = 0$ (solid) and $w = 0.5$ m/s (dashed); the curves should coincide. (b) Ground-frame altitude $z_0(t)$; the slopes differ by exactly w . (c) Both runs started at rest in the ground frame: maximum shape difference and tip-radius difference, normalized by L ; both decay exponentially with time constant 0.24 s.

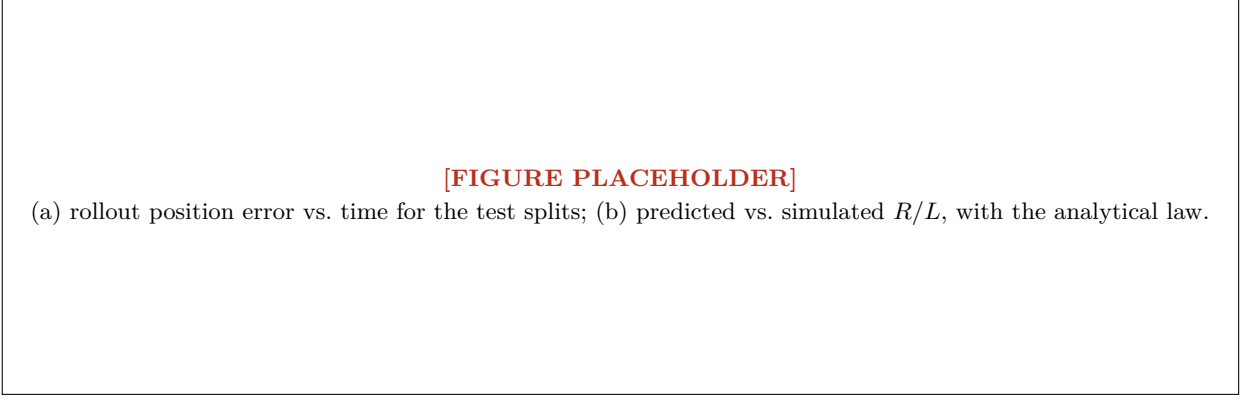


FIG. 9. Surrogate performance. (a) Rollout error of the node positions versus time, for interpolation and unseen- N test sets, with and without long-range message passing. (b) Steady-state tip radius predicted by the surrogate versus the DER simulation and the analytical law of Eq. (28).

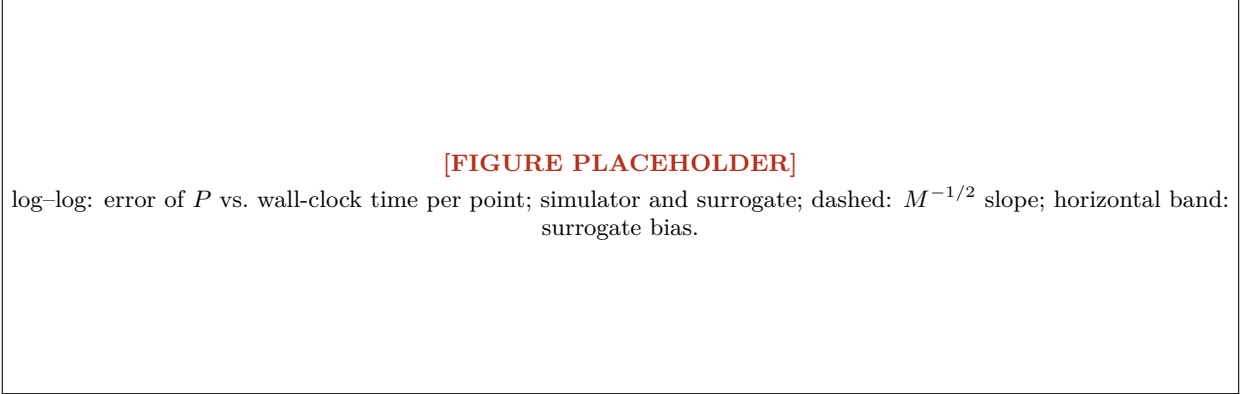


FIG. 10. Accuracy versus computational cost for the estimate of the take-off probability ($N = ?$, $\bar{F}_l = ?$, $\sigma_w = ?$ m/s). Mean absolute error $|P - P_{\text{ref}}|$ as a function of wall-clock time per point, for the simulator (circles) and the surrogate (squares), obtained by varying the number of realizations M . The dashed line has slope $-1/2$ (statistical error). The error of the surrogate saturates at its systematic bias; the crossing of the two curves marks the budget above which the simulator is preferable.

VIII. DISCUSSION

This section is written in advance as a frame. Statements marked with ? depend on the numerical results and must be confirmed or corrected once Figs. 2–8 are available. Alex and Vlad should rewrite it in their own words.

A. What this work adds

Compared with [1], the present work (i) places the charged multi-thread bundle in realistic near-surface turbulence and treats take-off as a stochastic event; (ii) gives an analytical drift–diffusion reference, Eq. (17), in which the electrostatic lift enters through $\bar{F}_l$ and the turbulence through a Péclet number; and (iii) provides an exact first integral and a closed-form opening law, $R/L \propto (S_N \Lambda)^{1/3}$, for tip-charged threads. Turbulent ballooning has been simulated before only for single, uncharged filaments [17, 18]. Earlier work treated the two lift mechanisms mostly separately: aerodynamic models without charge [7, 8], and electrostatic models in still air [1, 2]. Field and laboratory studies indicate that both play a role [9, 10]. Equation (17) makes this quantitative: electrostatic lift shifts the drift U_0 , while turbulence sets the width of the transition. **[TO DO: State by how much the simulated P departs from Eq. (17), for which N and ℓ/L , and whether multiple threads increase or decrease P at fixed $\bar{F}_l$ (Fig. 6).]**

B. What the learned surrogate adds

[TO DO: Octavian: interpret the results of Sec. VII C. Does the surrogate respect the analytical limits (first integral, opening law, threshold)? Where does it fail (large Λ , entanglement, unseen N)? What speed-up does it offer for parameter sweeps and inverse problems, given the break-even Q^* and the bias found in Sec. VII C 1?]

C. Order-of-magnitude assessment of the mechanism

Over flat ground ($E_0 \approx 120$ V/m), a spider of 1 mg carrying a total charge of 1 nC receives an electrostatic force of $\sim 1.2 \times 10^{-7}$ N, about 1% of its weight ($\sim 10^{-5}$ N). Electrostatic lift alone therefore requires either a field enhanced by a prominence (grass tips, twigs), where E reaches 10^4 – 10^5 V/m [1, 3], or charges about two orders of magnitude larger than those measured. In the language of the model, take-off from flat ground is essentially aerodynamic ($\bar{F}_l \ll 1$, w_c close to the settling speed), whereas take-off from an elevated point can be electrostatic or mixed. As the spider moves away from the prominence, E decays towards E_0 and $\bar{F}_l$ drops. Electrostatic forces are therefore most relevant *at take-off and in the first centimetres to metres of flight*, and much less during long-distance transport. **[TO DO: Confirm with a simulation in which the spider leaves the enhanced-field region described by Eq. (2).]**

D. Potential applications

- **Dispersal ecology.** Combined with routinely measured meteorological quantities (u_* , and hence σ_w ; atmospheric potential gradient), the take-off probability predicts under which weather conditions ballooning is likely, and how it depends on spider mass and number of threads.
- **Charged filament bundles.** The opening law (28) applies to any set of thin charged filaments in a viscous medium, for example electrospinning jets, charged fibres in filtration, and electrostatically interacting hair or fibre arrays [12].
- **Passive bio-inspired fliers.** The analysis gives design rules for ultralight passive fliers or airborne sensors lifted by drag on filaments, with or without charge. The number and length of the filaments set the sink speed, and hence the updraft required to stay aloft.
- **Atmospheric electricity.** The opening of a charged bundle scales as $E^{-1/3}$ [Eq. (28)], which suggests a passive indicator of the local field. The weak exponent and the sensitivity to charge leakage make this a qualitative rather than a practical measurement principle.
- **Benchmark for learned simulators.** The simulations form a controlled data set with known analytical limits, suitable for testing graph-based surrogate models of filament dynamics with long-range interactions [13, 14].

E. Limitations

a. Of the model.

- RFT neglects hydrodynamic interactions between threads, although the threads are very close near the spider [1]. A slender-body treatment would reduce the drag of the bundle near the spider.
- The charge distribution along the thread is unknown; only two limiting cases are treated.
- The charge is constant in time. Silk conductivity increases roughly exponentially with relative humidity [11], so in humid air the charge may leak within the flight time.
- The turbulence is synthetic, homogeneous, and isotropic, with statistics matched only for the vertical component; the growth of the eddy size with height, mean shear, and convective plumes are not included.
- Values of w_c above ~ 1 m/s (Table II) lie outside the range of validity of the low-Reynolds-number drag laws.
- The drift-diffusion reference (17) assumes a height-independent diffusivity; near the ground K grows with z .

b. *Of the mechanism.* The electrostatic mechanism depends on field enhancement near objects and on nanocoulomb charges whose origin (triboelectric charging during spinning, or induction) is not established. The measured charges [3] are indirect, inferred from the motion in a known field. The mechanism thus explains take-off in calm air from elevated points, but it cannot on its own explain long-distance transport at altitudes of hundreds of metres.

F. Future work

1. Anisotropic, height-dependent surface-layer turbulence (e.g. from large-eddy simulation), mean shear, and convective plumes.
2. Time-dependent charge, with a leakage rate depending on humidity [11].
3. An analytical or asymptotic solution for uniformly charged threads.
4. Hydrodynamic interactions between threads (slender-body theory or regularized Stokeslets).
5. Comparison with laboratory measurements of the opening angle versus applied field.
6. Learned surrogate simulators (graph neural networks with long-range message passing) trained on the present data, tested on unseen N and parameters, and checked against the analytical results of Secs. III–IV.

IX. CONCLUSIONS

We placed the charged multi-thread ballooning model of Habchi and Jawed [1] in synthetic near-surface turbulence with surface-layer statistics. A uniform flow leaves the bundle unchanged, so the relevant effect of the air is its spatial and temporal variability on the scale of the bundle. Take-off then becomes a stochastic event; in the limit of large, slow eddies its probability follows a drift-diffusion law that smooths the deterministic threshold $\bar{F}_l = 1$ into a sigmoid set by a Péclet number. For tip-charged threads we derived the exact first integral $T^\beta \sin \theta = \text{const.}$ and the closed-form opening law $R/L = [S_N \Lambda g(\tau)/4]^{1/3}$. The simulations reproduce the validation of [1] within 1%, confirm the opening law within 1% for $\Lambda < 1$, and show that the take-off probability departs from the drift-diffusion law by up to 10% for $N = 8$ and $\ell/L = 1$. An order-of-magnitude assessment indicates that electrostatic lift matters at take-off from elevated points, whereas over flat ground the aerodynamic mechanism dominates. Finally, a graph neural network surrogate trained on bundles with $N \leq 8$ threads predicts 10% for $N = 8$ with an error of 10%, at a cost 10 times lower than the full simulation.

ACKNOWLEDGMENTS

[TO DO:]

TABLE XI. HDF5 layout: one file per run.

Dataset / attribute	Shape	Content
/params (attributes)	—	N, N_t, L, r, m, r_s, q , charge model, Y, μ , field model, z_0, h
/turb (attributes, arrays)	$(N_k, \cdot)$	$\sigma_w, \ell, U_h, \lambda$, seed; $\mathbf{k}_n, \mathbf{a}_n, \mathbf{b}_n, \omega_n$
/t	(n_t)	output times (s)
/x	$(n_t, NN_t + 1, 3)$	node positions (m); node 0 = spider
/v	$(n_t, NN_t + 1, 3)$	node velocities (m/s)
/edges	$(NN_t, 2)$	node connectivity
/thread_id	$(NN_t + 1)$	thread index per node (−1 for spider)
/q_node	$(NN_t + 1)$	nodal charges (C)
/outcome (attributes)	—	rise/fall/timeout, exit time, entangled flag; still-air V, R, θ_L

AUTHOR CONTRIBUTIONS

[TO DO: .]

DATA AVAILABILITY

[TO DO: .]

Appendix A: Resistive force coefficients

For a slender filament of length L and radius $r \ll L$ at low Reynolds number, the drag coefficients per unit length of Eq. (4) follow from slender-body theory to leading order in $1/\ln(L/r)$. **[TO DO: Give the reference used and check the validity condition on the thread Reynolds number, $\text{Re} = \rho_{\text{air}}|\mathbf{v} - \mathbf{u}|r/\mu$, for the velocities in Table II.]**

Appendix B: Derivation of the opening law

With $T(s) = T_0 + \eta_{\parallel} U s$ and $\theta(s) = \theta_L [T_L/T(s)]^{\beta}$,

$$R = \int_0^L \theta \, ds = \frac{\theta_L T_L^{\beta}}{\eta_{\parallel} U} \int_{T_0}^{T_L} T^{-\beta} \, dT = \theta_L L \frac{\tau^{1-\beta} - 1}{(\beta - 1)(1 - \tau)}, \quad (\text{B1})$$

using $\eta_{\parallel} UL = T_L(1 - \tau)$. Combining with $\theta_L = F_C/(qE)$ and Eq. (22) gives Eq. (28). **[TO DO: Alex and Vlad: add the first correction in θ^2 and estimate the error for $\Lambda \sim 10^{-2}$.]**

Appendix C: Simulation parameters

Appendix D: List of symbols

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TABLE XII. Numerical parameters used in all simulations.

Parameter	Symbol	Value
Edges per thread	N_t	100 (convergence study: 25–200)
Initial / maximum / minimum time step	$\Delta t_0, \Delta t_{\max}, \Delta t_{\min}$	10^{-4} s, 10^{-2} s, 10^{-9} s
Newton tolerance	ϵ	10^{-10} N (on $\ \mathbf{F}\ _1$)
Maximum Newton iterations	K	20
Steady-state tolerance and window	δ, t_w	10^{-6} , 0.1 s
Initial thread spacing	d_0	100 μm
Poisson ratio; twist inertia	ν ; —	0.5; $\rho_t J l_0 / 2$
Tangent dependence of $\mathbf{C}$ in $\mathbf{J}$	—	lagged
Output interval	Δt_{out}	10^{-2} s (10^{-3} s for test 1)
Code and version	—	Python 3.14, NumPy 2.5, SciPy 1.18; commit 88246c2
Hardware	—	Apple M4 Pro, 10 performance cores, CPU only

TABLE XIII. Main symbols.

N	number of threads	$\eta_{\parallel}, \eta_{\perp}$	RFT drag coefficients
L, r	thread length, radius	β	$\eta_{\perp} / \eta_{\parallel}$
m, r_s	spider mass, radius	ζ_s	Stokes drag coefficient of the spider
q	charge per thread	$T(s)$	tension
$E(z)$	atmospheric electric field	$\theta(s)$	angle from the vertical
$\mathbf{u}, w$	air velocity, updraft speed	R	radius of the tip circle
U	velocity relative to air	S_N	Coulomb sum, Eq. (22)
V	ground-frame velocity	τ	tension ratio T_0/T_L
$\bar{F}_l$	lift-to-weight ratio	Λ	Coulomb-to-lift ratio
$\bar{w}$	dimensionless updraft	w_c, w_s	mean-flow threshold, settling speed
σ_w, ℓ	turbulence intensity, scale	P, K	take-off probability, diffusivity

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